

Safety Data Sheet

according to HPR, Schedule 1

Printing date 01/15/2021

Reviewed on 07/31/2020

1 Identification

- **Product identifier**
- **Trade name:** *Wash Reagent 2*
- **Catalog or product number:** 10005487, 4307591
- **Application of the substance / the mixture** Laboratory chemicals
- **Details of the supplier of the safety data sheet**
- **Manufacturer/Supplier:**
Bio-Rad Laboratories (Canada) Ltd.
1329 Meyerside Drive
Mississauga, Ontario
L5T 1C9
Canada
- **Information department:** sales_canada@bio-rad.com
- **Emergency telephone number:** 1-800-424-6723 option 2

phone: 1-800-268-0213 Fax: 1-888-913-9779

2 Hazard identification

· Classification of the substance or mixture

Flammable Liquids - Category 2

Acute Toxicity (Dermal) – Category 4

Skin Irritation - Category 2

Serious Eye Damage - Category 1

Carcinogenicity – Category 2

Reproductive Toxicity - Category 2

Specific Target Organ Toxicity - Single Exposure - Category 3

Specific Target Organ Toxicity - Repeated Exposure -
Category 1

H225 Highly flammable liquid and vapour.

H312 Harmful in contact with skin.

H315 Causes skin irritation.

H318 Causes serious eye damage.

H351 Suspected of causing cancer.

H361 Suspected of damaging fertility or the unborn child.

H336 May cause drowsiness or dizziness.

H372 Causes damage to organs through prolonged or
repeated exposure.

· Label elements

· **GHS label elements** The product is classified and labeled according to the Globally Harmonized System (GHS).

· Hazard pictograms



GHS02 GHS05 GHS07 GHS08

· Signal word

Danger

· Hazard-determining components of labeling:

trichloromethane

acetone

hydrochloric acid

propan-2-ol

· Hazard statements

H225 Highly flammable liquid and vapour.

H312 Harmful in contact with skin.

H315 Causes skin irritation.

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*H318 Causes serious eye damage.**H351 Suspected of causing cancer.**H361 Suspected of damaging fertility or the unborn child.**H336 May cause drowsiness or dizziness.**H372 Causes damage to organs through prolonged or repeated exposure.***Precautionary statements***P303+P361+P353 IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water [or shower].**P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.**P310 Immediately call a poison center/doctor.**P362+P364 Take off contaminated clothing and wash it before reuse.**P322 Specific measures (see on this label).**P405 Store locked up.**P501 Dispose of contents/container in accordance with local/regional/national/international regulations.*

3 Composition/Information on ingredients

Chemical characterization: Mixtures**Description:** Mixture of the substances listed below with non-hazardous additions.**Listing of dangerous and non-hazardous components:**

67-64-1	acetone Flammable Liquids - Category 2, H225 Eye Irritation - Category 2A, H319; Specific Target Organ Toxicity - Single Exposure - Category 3, H336	60-80% w/w *
67-63-0	propan-2-ol Flammable Liquids - Category 2, H225 Eye Irritation - Category 2A, H319; Specific Target Organ Toxicity - Single Exposure - Category 3, H336	1-5% w/w *
67-66-3	trichloromethane Flammable Liquids - Category 2, H225 Acute Toxicity (Dermal) - Category 2, H310; Acute Toxicity (Inhalation) - Category 3, H331 Carcinogenicity - Category 2, H351; Reproductive Toxicity - Category 2, H361; Specific Target Organ Toxicity - Repeated Exposure - Category 1, H372 Acute Toxicity (Oral) - Category 4, H302; Skin Irritation - Category 2, H315; Eye Irritation - Category 2A, H319	1-5% w/w *
7647-01-0	hydrochloric acid Skin Corrosion - Category 1B, H314; Serious Eye Damage - Category 1, H318 Specific Target Organ Toxicity - Single Exposure - Category 3, H335	3-<5% w/w *
	Component 7 Flammable Liquids - Category 3, H226 Acute Toxicity (Oral) - Category 4, H302	1-5% w/w *

* Actual concentration ranges are withheld as a trade secret.

Additional information For the wording of the listed hazard phrases refer to section 16.

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4 First-aid measures

- **Description of first aid measures**

- **General information**

Immediately remove any clothing soiled by the product.

Symptoms of poisoning may even occur after several hours; therefore medical observation for at least 48 hours after the accident.

- **After inhalation** Supply fresh air; consult doctor in case of complaints.

- **After skin contact**

Generally the product does not irritate the skin.

Immediately wash with water and soap and rinse thoroughly.

Immediately rinse with water.

- **After eye contact** Rinse opened eye for several minutes under running water. Then consult a doctor.

- **After swallowing** Rinse mouth with water. Seek medical attention and appropriate follow-up.

- **Information for doctor**

- **Most important symptoms and effects, both acute and delayed** No further relevant information available.

- **Indication of any immediate medical attention and special treatment needed** No further relevant information available.

5 Fire-fighting measures

- **Extinguishing media**

- **Suitable extinguishing agents**

CO₂, extinguishing powder or water spray. Fight larger fires with water spray or alcohol resistant foam.

- **For safety reasons unsuitable extinguishing agents** Water with full jet.

- **Special hazards arising from the substance or mixture** During heating or in case of fire poisonous gases are produced.

- **Advice for firefighters**

- **Protective equipment:**

Mount respiratory protective device.

Wear self-contained respiratory protective device.

6 Accidental release measures

- **Personal precautions, protective equipment and emergency procedures**

Mount respiratory protective device.

Wear protective equipment (See section 8). Keep unprotected persons away.

Wear protective clothing.

- **Environmental precautions:**

Prevent seepage into sewage system, workpits and cellars.

Do not allow to enter sewers/ surface or ground water.

- **Methods and material for containment and cleaning up:**

Absorb with liquid-binding material (sand, diatomite, acid binders, universal binders, sawdust).

Use neutralizing agent.

Dispose contaminated material as waste according to item 13.

Ensure adequate ventilation.

- **Reference to other sections**

See Section 7 for information on safe handling

See Section 8 for information on personal protection equipment.

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See Section 13 for disposal information.

7 Handling and storage

- **Handling**
- **Precautions for safe handling**
Ensure good ventilation/exhaust at the workplace.
Open and handle receptacle with care.
Prevent formation of aerosols.
- **Information about protection against explosions and fires:**
Keep ignition sources away - Do not smoke.
Protect against electrostatic charges.
Keep respiratory protective device available.
- **Conditions for safe storage, including any incompatibilities**
- **Storage**
- **Requirements to be met by storerooms and receptacles:** Store in a cool location.
- **Information about storage in one common storage facility:** Not required.
- **Further information about storage conditions:**
Keep receptacle tightly sealed.
Store in cool, dry conditions in well sealed receptacles.
- **Specific end use(s)** No further relevant information available.

8 Exposure controls/ Personal protection

- **Additional information about design of technical systems:** No further data; see item 7.
- **Control parameters**

· **Components with limit values that require monitoring at the workplace:**

67-64-1 acetone

EL STEL: 500 ppm
TWA: 250 ppm

EV STEL: 750 ppm
TWA: 500 ppm

67-63-0 propan-2-ol

EL STEL: 400 ppm
TWA: 200 ppm

EV STEL: 400 ppm
TWA: 200 ppm

67-66-3 trichloromethane

EL TWA: 2 ppm
IARC 2B; R

EV TWA: 49 mg/m³, 10 ppm

7647-01-0 hydrochloric acid

EL Ceiling: 2 ppm

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- EV Ceiling: 2 ppm
- **Additional information:** The lists that were valid during the creation were used as basis.
 - **Exposure controls**
 - **Personal protective equipment**
 - **General protective and hygienic measures**
 Keep away from foodstuffs, beverages and feed.
 Immediately remove all soiled and contaminated clothing
 Wash hands before breaks and at the end of work.
 Store protective clothing separately.
 Avoid contact with the skin.
 Avoid contact with the eyes.
 Avoid contact with the eyes and skin.
 - **Breathing equipment:**
 In case of brief exposure or low pollution use respiratory filter device. In case of intensive or longer exposure use respiratory protective device that is independent of circulating air.
 Not necessary if room is well-ventilated. Adequate ventilation required.
 - **Protection of hands:** Protective gloves.
 - **Material of gloves** Synthetic gloves
 - **Penetration time of glove material**
 The exact break through time has to be found out by the manufacturer of the protective gloves and has to be observed.
 - **Eye protection:**
 Safety glasses
 Tightly sealed goggles.

9 Physical and chemical properties

· Information on basic physical and chemical properties

· General Information

· Appearance:

Form:	Liquid
Color:	Colorless
Odor:	Recognizable
Odor threshold:	Not determined.

· pH-value at 20 °C: 5.5

· Change in condition

Melting point/Melting range:	undetermined
Boiling point/Boiling range:	55 °C

· Flash point: < 0 °C

· Flammability (solid, gaseous) Not applicable.

· Ignition temperature: 340 °C

· Decomposition temperature: Not determined.

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· Auto igniting:	Product is not selfigniting.
· Danger of explosion:	Product is not explosive. However, formation of explosive air/vapor mixtures are possible.
· Explosion limits:	
Lower:	2.6 Vol %
Upper:	13 Vol %
· Vapor pressure at 20 °C:	233 hPa
· Density at 20 °C:	0.82281 g/cm ³
· Relative density	Not determined.
· Vapor density	Not determined.
· Evaporation rate	Not determined.
· Solubility in / Miscibility with Water:	Fully miscible
· Partition coefficient (n-octanol/water):	Not determined.
· Viscosity:	
dynamic:	Not determined.
kinematic:	Not determined.
· Solvent content:	
Organic solvents:	90.0 %
· Other information	No further relevant information available.

10 Stability and reactivity

- **Reactivity** No further relevant information available.
- **Chemical stability**
- **Thermal decomposition / conditions to be avoided:** No decomposition if used according to specifications.
- **Possibility of hazardous reactions** No dangerous reactions known
- **Conditions to avoid** No further relevant information available.
- **Incompatible materials:** No further relevant information available.
- **Hazardous decomposition products:** No dangerous decomposition products known

11 Toxicological information

- **Information on toxicological effects**
- **Acute toxicity:**

· **LD/LC50 values for hazardous components per OSHA criteria:**

67-64-1 acetone

Oral	LD50	5,800 mg/kg (rat)
Dermal	LD50	20,000 mg/kg (rbt)

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67-63-0 propan-2-ol

Oral	LD50	4,570 mg/kg (rat)
Dermal	LD50	13,400 mg/kg (Rabbit)
Inhalative	LC50/4 h	30 mg/l (rat)

67-66-3 trichloromethane

Oral	LD50	908 mg/kg (rat)
Dermal	LD50	75 mg/kg (rat)

Component 7

Oral	LD50	1,300 mg/kg (rat)
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· **Primary irritant effect:**· **on the skin:** Irritant to skin and mucous membranes.· **on the eye:**

Strong caustic effect.

Strong irritant with the danger of severe eye injury.

Irritant effect.

· **Sensitization:** No sensitizing effects known.· **Additional toxicological information:**

The product shows the following dangers according to the calculation method of the General EU Classification Guidelines for Preparations as issued in the latest version:

Harmful

Irritant

· **Carcinogenic categories**· **IARC (International Agency for Research on Cancer)**

67-63-0	propan-2-ol	3
67-66-3	trichloromethane	2B
7647-01-0	hydrochloric acid	3

· **NTP (National Toxicology Program)**

67-66-3	trichloromethane	R
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12 Ecological information

· **Toxicity**· **Aquatic toxicity:** No further relevant information available.· **Persistence and degradability** No further relevant information available.· **Behavior in environmental systems:**· **Bioaccumulative potential** No further relevant information available.· **Mobility in soil** No further relevant information available.· **Additional ecological information:**· **General notes:**

Water danger class 3 (Internal calculation) (Self-assessment): extremely hazardous for water.

Do not allow product to reach ground water, water course or sewage system, even in small quantities.

Must not reach bodies of water or drainage ditch undiluted or unneutralized.

Danger to drinking water if even extremely small quantities leak into the ground.

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- **Results of PBT and vPvB assessment**
- **PBT:** Not applicable.
- **vPvB:** Not applicable.
- **Other adverse effects** No further relevant information available.

13 Disposal considerations

- **Waste treatment methods**
- **Recommendation**
Must not be disposed of together with household garbage. Do not allow product to reach sewage system.
Dispose of waste in accordance to applicable national, regional, or local regulations.
- **Uncleaned packagings:**
- **Recommendation:** Disposal must be made according to official regulations.

14 Transport information

- | | |
|--|---|
| · UN-Number | UN1993 |
| · DOT/TDG, ADR, IMDG, IATA | |
| · UN proper shipping name | Flammable liquids, n.o.s. (Acetone, Chloroform) |
| · DOT/TDG | 1993 FLAMMABLE LIQUID, N.O.S. (vapour pressure at 50°C not more than 110 kPa) (ACETONE, CHLOROFORM) |
| · ADR | 1993 FLAMMABLE LIQUID, N.O.S. (vapor pressure at 50°C at most 110 kPa) (ACETONE, CHLOROFORM) |
| · IMDG, IATA | FLAMMABLE LIQUID, N.O.S. (ACETONE, CHLOROFORM) |
| · Transport hazard class(es) | |
| · DOT, ADR, IMDG, IATA | |
| · Class | 3 Flammable liquids |
| · Label | 3 |
| · Packing group | |
| · DOT/TDG, ADR, IMDG, IATA | II |
| · Environmental hazards: | |
| · Marine pollutant: | No |
| · Special precautions for user | Warning: Flammable liquids |
| · Hazard identification number (Kemler code): | 33 |
| · EMS Number: | F-E, S-E |
| · Segregation groups | Liquid halogenated hydrocarbons |
| · Stowage Category | B |
| · Transport in bulk according to Annex II of MARPOL73/78 and the IBC Code | Not applicable. |

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· **Transport/Additional information:**

· DOT/TDG

· Quantity limitations

Void

Void

· ADR

· Excepted quantities (EQ)

Code: E2

Maximum net quantity per inner packaging: 30 ml

Maximum net quantity per outer packaging: 500 ml

· IMDG

· Limited quantities (LQ)

· Excepted quantities (EQ)

1L

Code: E2

Maximum net quantity per inner packaging: 30 ml

Maximum net quantity per outer packaging: 500 ml

· UN "Model Regulation":

UN 1993 FLAMMABLE LIQUID, N.O.S. (VAPOUR
PRESSURE AT 50°C NOT MORE THAN 110 KPA)
(ACETONE, CHLOROFORM), 3, II

15 Regulatory information

· Safety, health and environmental regulations/legislation specific for the substance or mixture

· SARA (Superfund Amendments and Reauthorization Act of 1986 - USA)

· Section 302/304 (40CFR355.30 / 40CFR355.40):

67-66-3 trichloromethane

7647-01-0 hydrochloric acid

· Section 313 (40CFR372.65):

67-63-0 propan-2-ol

67-66-3 trichloromethane

7647-01-0 hydrochloric acid

· TSCA (Toxic Substances Control Act):

67-64-1 acetone

ACTIVE

67-63-0 propan-2-ol

ACTIVE

67-66-3 trichloromethane

ACTIVE

7647-01-0 hydrochloric acid

ACTIVE

Component 7

*

· MAK (German Maximum Workplace Concentration)

67-66-3 trichloromethane

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· National regulations

This product has been classified in accordance with the hazard criteria of the Controlled Products Regulations and the MSDS contains all of the information required by those regulations

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· **Technical instructions (air):**

Class	Share in %
I	1-5
III	1-5
NK	65-85

- **Water hazard class:** Water danger class 3 (Self-assessment): extremely hazardous for water.
- **Chemical safety assessment:** A Chemical Safety Assessment has not been carried out.

16 Other information

This information is based on our present knowledge. However, this shall not constitute a guarantee for any specific product features and shall not establish a legally valid contractual relationship.

- **Department issuing SDS:** Environmental Health and Safety.

· **Contact:**

Life Science Group, Environmental Health and Safety, 2000 Alfred Nobel Drive, Hercules, California, 94547: 1(510) 741-1000

Diagnostic Group, Environmental Health and Safety, 4000 Alfred Nobel Drive, Hercules, California, 94547: 1(510) 724-7000

- **Date of the latest revision of the safety data sheet** 01/15/2021 / -

· **Abbreviations and acronyms:**

IMDG: International Maritime Code for Dangerous Goods

DOT: US Department of Transportation

IATA: International Air Transport Association

EINECS: European Inventory of Existing Commercial Chemical Substances

ELINCS: European List of Notified Chemical Substances

CAS: Chemical Abstracts Service (division of the American Chemical Society)

LC50: Lethal concentration, 50 percent

LD50: Lethal dose, 50 percent

PBT: Persistent, Bioaccumulative and Toxic

vPvB: very Persistent and very Bioaccumulative

- *** Data compared to the previous version altered.**